

**RAMCO INSTITUTE OF TECHNOLOGY**  
**Department of Computer Science and Engineering**

**Academic Year: 2019- 2020 (Odd Semester)**

Innovative practice(s) description

**Course code and title:** CS8592 Object Oriented Analysis and Design

**Name of the Instructor(s):** Mr.D.Lakshmanan, AP(S.G)/CSE

**Date & Time:** 16.08.2019 & 9.50AM to 10.40AM

**Date: 16.08.2019**

**Case Study**

**Topic- Case Study of Library Information System**

**Objectives**

The primary goal of the case study is to give users an opportunity to see how the image guidelines apply in the context of an actual lab and theory topics.

**Justification for choosing this topic:**

This virtual lab is designed and implemented by IIT karagpur, India. The SE VLabs Institute has been recently setup to provide state-of-the-art research facilities in the field of Software Engineering. Activity diagram has explained clearly with the use of case study of a Library Information System for SE VLabs Institute. The activity diagram explained in this case study used various symbols such as Initial node, Action node, Decision, Merge, Fork, join and swimlane. By means of understanding this case study available in the virtual lab web site, students can draw activity diagram for other system as well.

The following link is used to access the virtual lab.

[http://vlabs.iitkgp.ernet.in/se/6/case\\_study/](http://vlabs.iitkgp.ernet.in/se/6/case_study/)

**Activity**

As the size and capacity of the institute is increasing with the time, it has been proposed to develop a Library Information System (LIS) for the benefit of students and employees of the institute. LIS will enable the members to borrow a book (or return it) with ease while sitting at his desk/chamber. The system also enables a member to extend the date of his borrowing if no other booking for that particular book has been made. For the library staff, this system aids them to easily handle day-to-day book transactions. The librarian, who has administrative privileges and complete control over the system, can enter a new record into the system when a new book has been purchased, or remove a record in case any book is

taken off the shelf. Any non-member is free to use this system to browse/search books online. However, issuing or returning books is restricted to valid users (members) of LIS only.

From the given problem we can identify minimum three different functionalities offered by the systems are:

1. Register a new member
2. Issue book
3. Update inventory

The above three activities diagrams are explained in the class rooms are as follows.

### 1.Activity diagram for registering a new member

To begin with, let's consider the activity diagram for user registration, as shown in figure - 01. below.

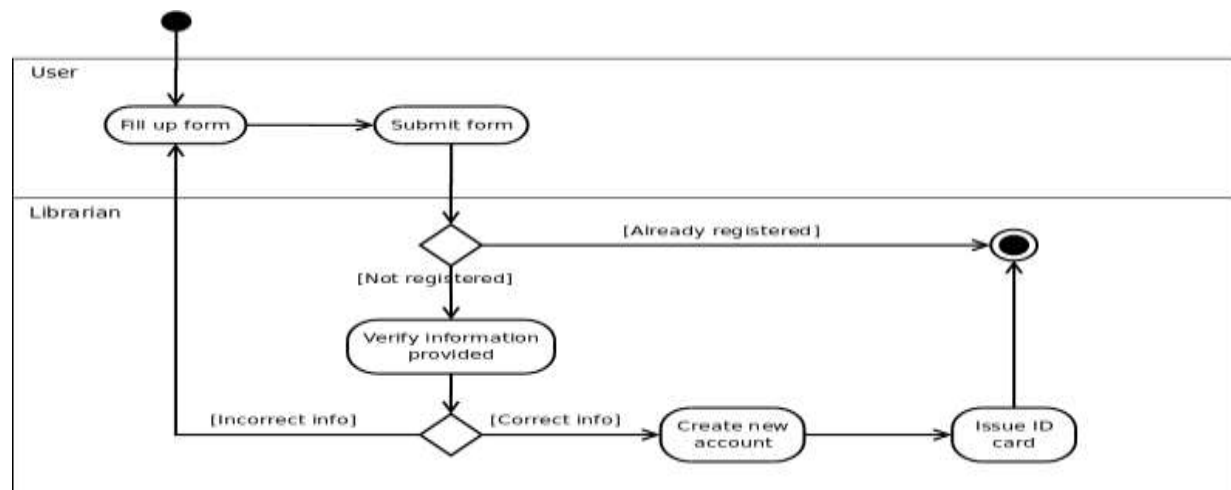


Figure-01: Activity diagram for new user registration

A new user fills up the registration form for library membership (either online or in paper), and submits to the librarian. Of course, an already registered user can't create another account for himself (or, herself). For users' who don't have an account already and have submitted their registration forms, the librarian verifies the information provided, possibly against the central database used by the institution. If all information have been provided correctly, librarian goes on with creating a new account for the user. Otherwise, the user is asked to provide all and correct information in his (her) registration form. Once a new account has been created for the user, he (she) is being issued an ID card, which is to be provided for any future transaction in the library.

Note that in the above diagram two swim lanes haven been shown indicated by the labels User and Librarian. The activities have been placed in swim lanes that correspond to the relevant role.

## 2. Activity diagram for issuing books

One of the major events that occur in any library is issue of books to it's members. Figure-02 tries to depict the workflow involved while issuing books.

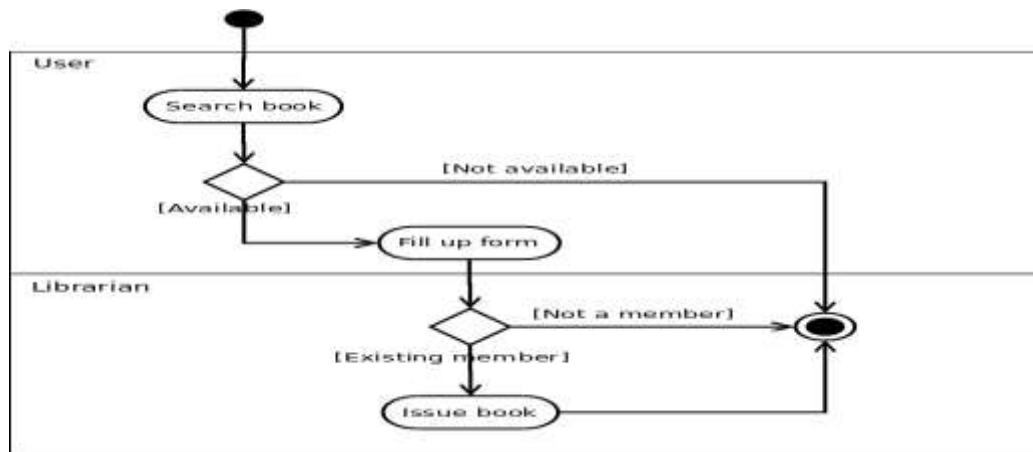


Figure-02: Activity diagram for issuing books

## 3. Activity Diagram for updating inventory

Now let's focus on figure-03, which shows the typical workflow of inventory update by the librarian. Note that since these are the tasks performed only by the librarian (and no one else plays a role), we skip the swim lanes.

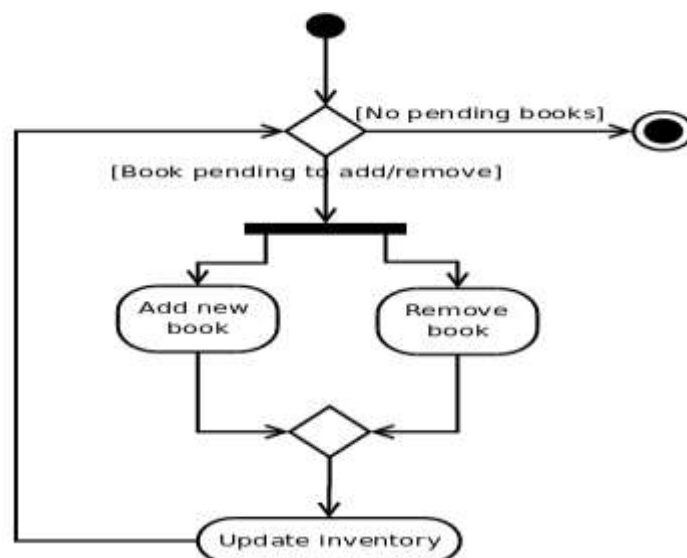


Figure-03: Activity diagram for updating inventory

Addition of new books and removing records of books taken off from the shelves could be done parallel. This means, one doesn't have to complete the task of addition of all new books before doing any removal. Merging of these two activities and the subsequent Update inventory activity indicates that it is not required to complete all addition and removals before proceeding to update the database. That is, a few books could be added, and then update the database, then again continues with the tasks.

Finally, the workflow terminates when all addition and removal tasks have been completed.

During the activity some of the students asked questions like

### **Frequent Ask Questions and answers.**

- 1. What is the need of swim lane?**
- 2. Why we want to use fork and join symbol?**

### **Answers**

- 1. What is the need of swim lane?**

A swim lane is a visual element used in process Activity diagram, or flowcharts that visually distinguishes job sharing and responsibilities for sub-processes of a business process. Swim lanes may be arranged either horizontally or vertically.

- 2. Why we want to use fork and join symbol?**

Fork - A fork has one flow entering and several exiting. A fork denotes that several processes are occurring in parallel.

Join - A join has several flows entering it and one exiting it. A join denotes that multiple parallel flows are merging at a single point. All flows going into the join must be completed before the next activity can start.

### **Reference:**

The above case study is referred from the Indian Institute of Technology Delhi virtual lab site below.

[http://vlabs.iitkgp.ernet.in/se/6/case\\_study/](http://vlabs.iitkgp.ernet.in/se/6/case_study/)

This activity helps in attaining the following CO – PO mapping:

| Course Outcome / Programme Outcome                                    | PO1 | PO2 | PO3 | PO6 |
|-----------------------------------------------------------------------|-----|-----|-----|-----|
| Express software design with UML dynamic and implementation diagrams. | 2   | 2   | 3   | 1   |

#### **SESSION OUTCOMES:**

- This makes the students to draw the activity diagram for any real time system by learning the case study of library management system.
- It provides opportunities for the students to acquire knowledge about real time systems activities.
- .

#### **Reflection Report**

- ❖ Students have got clear fundamental knowledge on activity diagram.
- ❖ Students worked joyfully because of visualization blow in their mind.
- ❖ I assisted to slow learners to draw the activity diagram on their laptop system using Argo UML tool.
- ❖ All the students have given good feedback and some of the students used this activity diagram for their Object Oriented Analysis and Design lab project.